Lab Report
CSE 302 – Computer Organization & Architecture

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Task#1: The function **`islower`**, shown in Figure 5.1, tests whether a character **`ch`** is lowercase or not. Write the **`main`** function of a program that reads a character **`ch`**, calls the function **`islower`**, and then prints a message to indicate whether **`ch`** is a lowercase character or not.

```
.data
msg1: .asciiz "Enter a character: "
msg_lower: .asciiz "The character is lowercase.\n"
msg_notlower: .asciiz "The character is NOT lowercase.\n"
.text
.globl main
main:
    li $v0, 4
    la $a0, msg1
    syscall
    li $v0, 12
    syscall
    move $t0, $v0
    li $t1, 97
    li $t2, 122
    blt $t0, $t1, not_lower
    bgt $t0, $t2, not_lower
    li $v0, 4
    la $a0, msg_lower
    syscall
    j end_prog
not_lower:
```

```
    li $v0, 4
    la $a0, msg_notlower
    syscall
end_prog:
    li $v0, 10
    syscall
```

Task#2: The recursive function **`fib(n)`** computes the n_{th} element in the Fibonacci sequence. Implement this function in MIPS. Write a **`main`** function to call **`fib`**.

```
.data
msg2: .asciiz "Enter n for Fibonacci: "
msg2_out: .asciiz "Fibonacci result: "
.text
.globl main
main:
    li $v0, 4
    la $a0, msg2
    syscall
    li $v0, 5       # read integer
    syscall
    move $a0, $v0  # n
    jal fib         # call fib(n)
    move $t0, $v0
    li $v0, 4
    la $a0, msg2_out
    syscall

    move $a0, $t0
```

```
    li $v0, 1
    syscall
    li $v0, 10
    syscall
fib:
    addi $sp, $sp, -8
    sw $ra, 4($sp)
    sw $a0, 0($sp)
    li $t0, 2
    blt $a0, $t0, fib_base
    lw $a0, 0($sp)
    addi $a0, $a0, -1
    jal fib
    move $t1, $v0
    lw $a0, 0($sp)
    addi $a0, $a0, -2
    jal fib
    add $v0, $v0, $t1
    j fib_end
fib_base:
    lw $a0, 0($sp)
    move $v0, $a0
fib_end:
    lw $ra, 4($sp)
    addi $sp, $sp, 8
    jr $ra
```